

Two-stage maximum Margin Preservation under Bounded Score Perturbations

Research Architecture Runtime
operator/two-stage-maximum

2026-07-25

Abstract

Two-stage maximum reduces to unique-maximizer Argmax margin stability: $\gamma > 2\varepsilon$ preserves the winner under $\|\delta\|_\infty \leq \varepsilon$. Lean: `LEAN.FULL`.

1 Theorems

Theorem 1 (Preservation). *Definitional reduction to Argmax margin invariance.*

Theorem 2 (Sharpness). *Definitional reduction to Argmax margin sharpness.*

2 Lean / certificate

`Research.Operators.TwoStageMaximum.Margin.`

3 Limitations

Mathlib $\mathbb{R}$ (`REAL_MATHLIB`); Argmax reduction.